江苏省苏州中学国际部
Suzhou High School – International Division

SZZX-ID
BP Physics

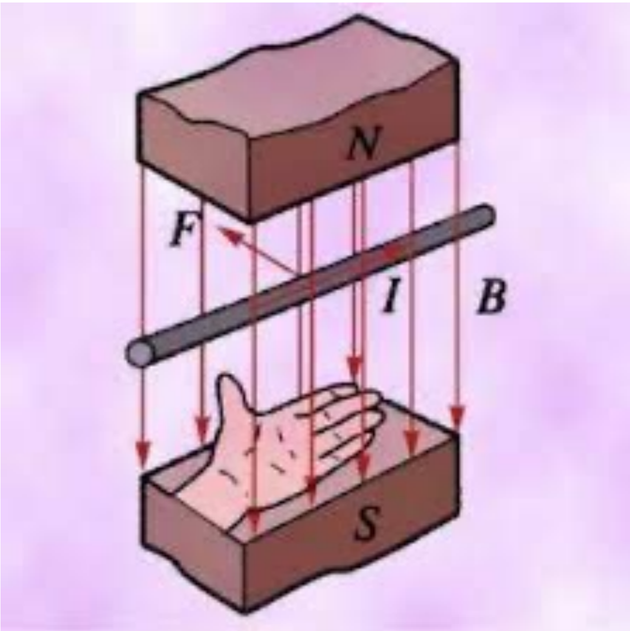
Motion of a charged particle in a magnetic field

Class Outline

Title of Unit	Magnetic field
Title of Lesson	Motion of a charged particle in a magnetic field
Starter Activity	Revision: use the left hand to determine the direction of induced magnetic force
Direct Instructions	Teaching Strategies: Demonstration, Lecture, Collaboration, Questioning, Inquiry
	- Questioning: revision – direction of magnetic force - Lecture: definition of B. charge particles doing circular motion in B field
	Differentiation: Content, Instructional Strategies, the Classroom, Products, Teacher
	- Students are advised to choose the tool (one of the two left hand rule) that they are comfortable with and practice.
	International Mindedness:
	Magnetic field of Earth is one of the very fundamental condition of earth being habitable
Lesson Objectives	By the end of the lesson, students will be able to: - understand the force of a moving charged particle in B field is qvBsinθ - able to derive the radius of the circular motion of a charged particle in B field - Understand how can E field and B field be combined to be a velocity selector
ATLs	Identify unit ATLs and how they are being addressed in the lesson
	Inquiry
IB Learner Profile	Link the lesson to the characteristics of the IB Learner attribute
	Inquirer, thinker
Teaching Activities	- MCQ: revision on the direction of force by using left-hand rule - Derive the unit of magnetic field
Plenary Session	Checking for understanding / Homework / Q & A / Reflection etc.
	Summarize that when v is perpendicular to B, charge moves in the magnetic field in a circular motion When the E and B field is combined, the undeflected particles are with fixed velocity

Revision - Determine the direction of magnetic force on a straight wire

Left-hand rule

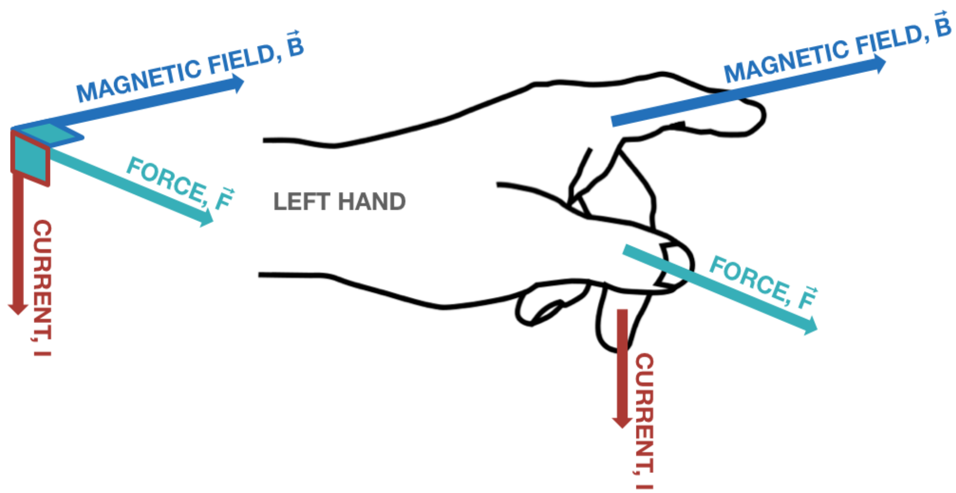


- Use left hand
- Let magnetic field lines go into the palm
- The four fingers pointing to the direction of current
- The direction of the thumb is the direction of magnetic force acting on the wire

Revision - Determine the direction of magnetic force on a straight wire – another way

Fleming's Left Hand Rule

The direction of the magnetic force can be determined using Fleming's Left Hand Rule.



All three fingers should be mutually perpendicular.

Warm up (revision) - 1

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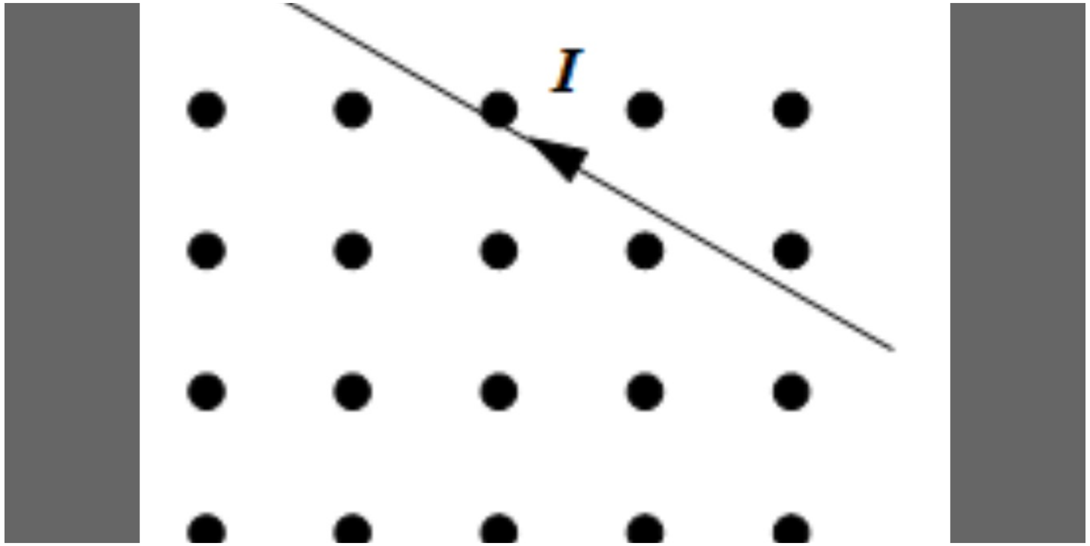
What is the direction of the magnetic force on the wire?



1. Top right cornor	2. Top left cornor	3. Bottom right cornor	4. Bottom left cornor

Warm up (revision) - 1

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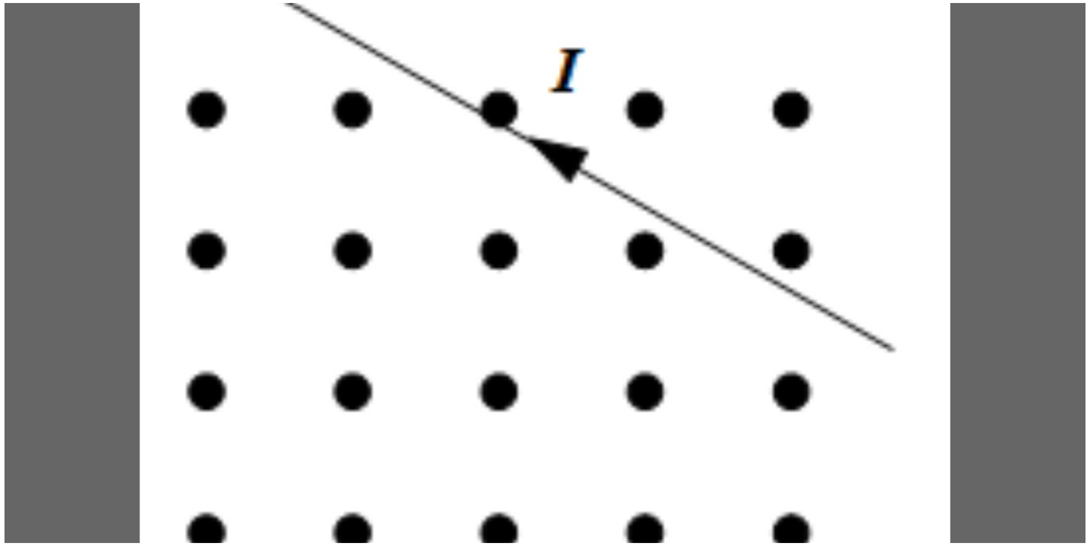
What is the direction of the magnetic force on the wire?



1. Top right cornor			

Warm up (revision) - 2

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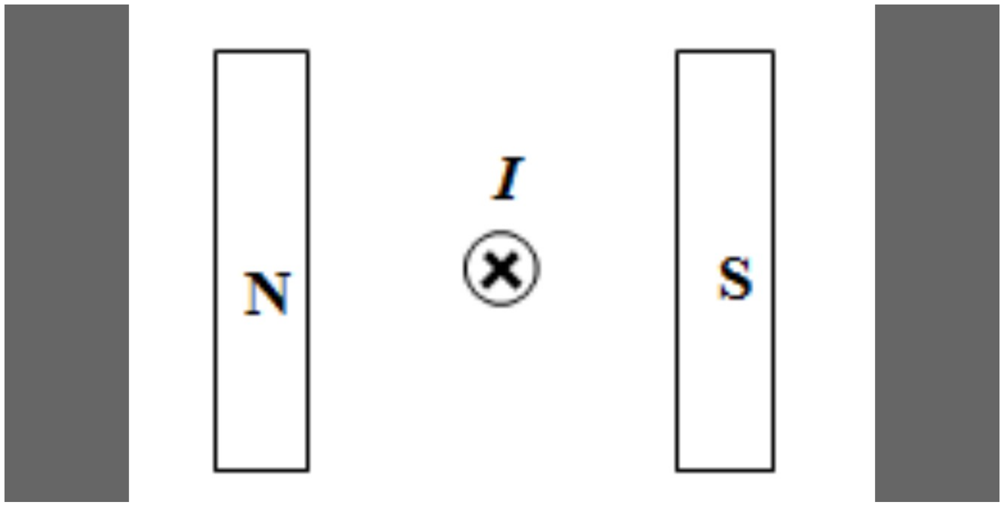
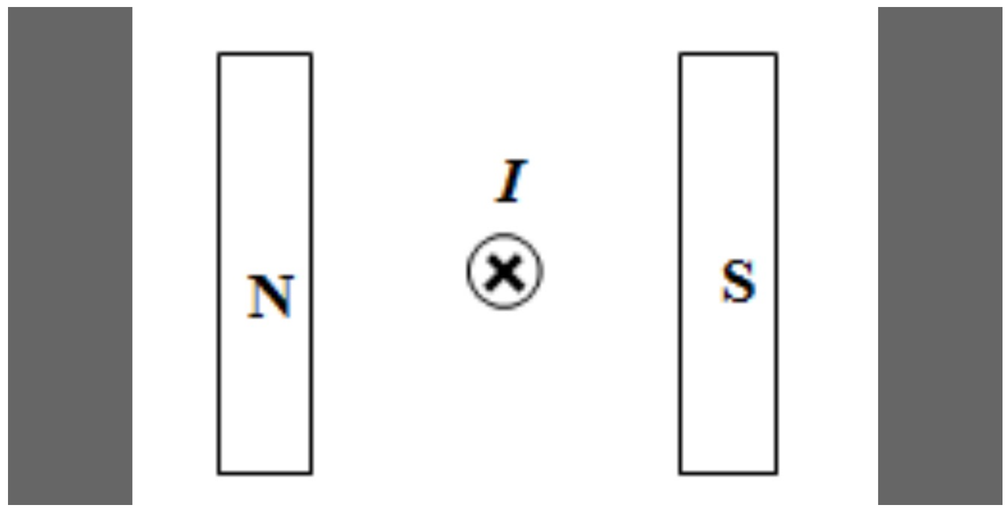
What is the direction of the magnetic force on the wire?



1. Left	2. Right	3. Up	4. Down

Warm up (revision) - 2

What is the direction of the magnetic force on the wire?



			4. Down

Warm up (revision) - 3

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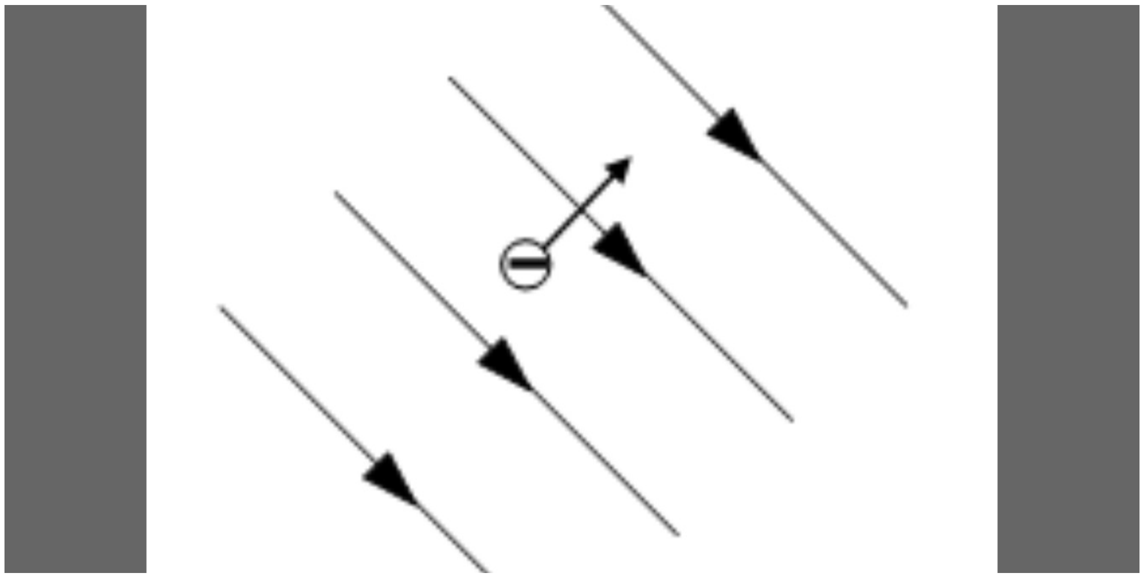
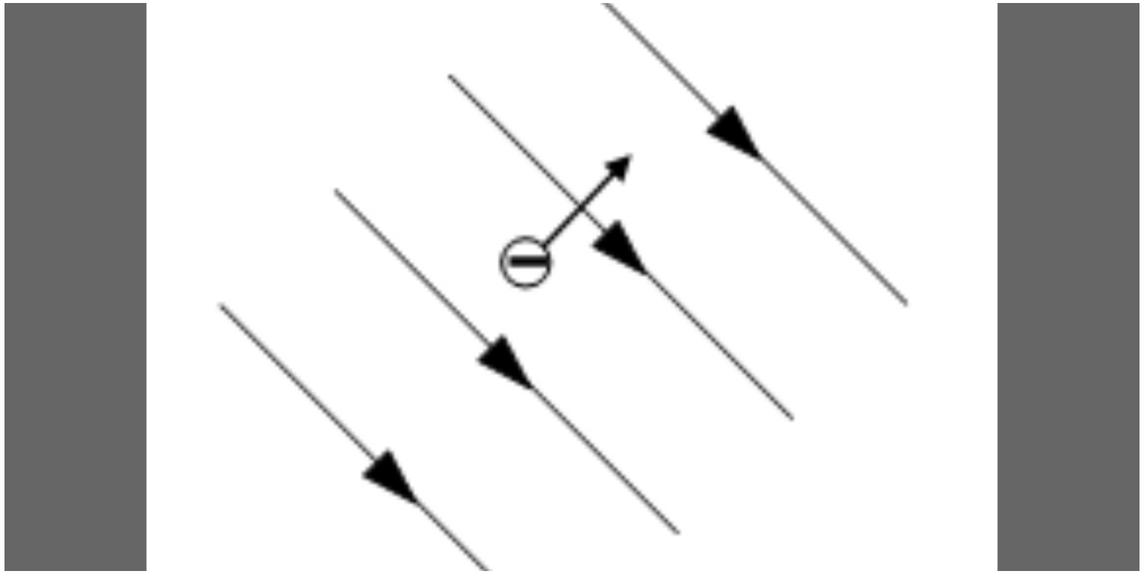
What is the direction of the magnetic force on the moving electron?



1. Up	2. Down	3. Into the screen	4. Out of the screen

Warm up (revision) - 3

What is the direction of the magnetic force on the moving electron?

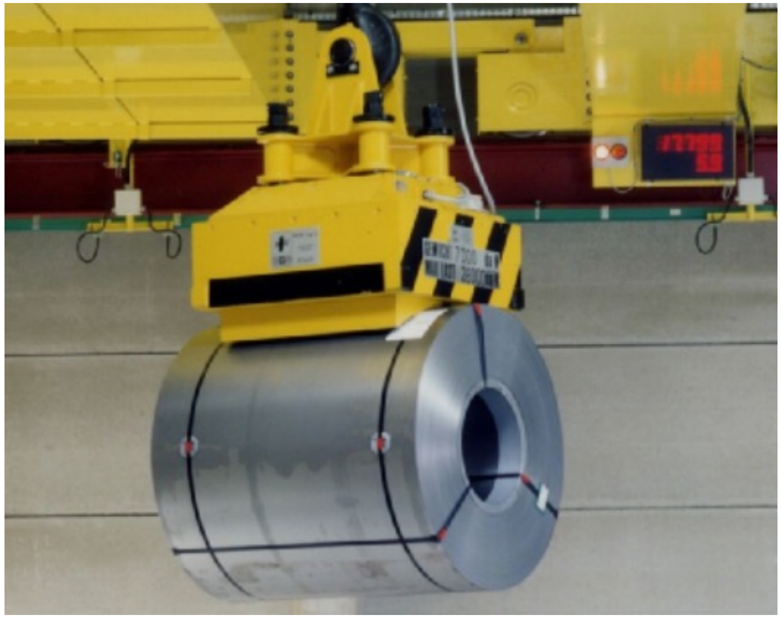


			4. Out of the screen

Quantify the strength of magnetic field

Weak magnetic field – attracts a few paper clips

Strong magnetic field – attracts tons of steel



How do we quantitatively describe the strength of a magnetic field?

Magnetic force and field strength

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- **Experiments show that an electric charge moving in a region of magnetic field experiences a new type of force called a magnetic force**
- **We can quantify the strength of the field by looking at the magnetic force of the field on the charge**
- **The magnitude of magnetic field, or magnetic flux density, B is defined as**

$$B = \frac{F}{qv\sin\theta}$$

- Unit of B: Tesla (T)
- A magnetic flux density of 1T produces a force of 1N on a charge of 1C moving at 1 ms^{-1} at right angle to the direction of the magnetic field

Qn: What is the unit Tesla in SI base units?

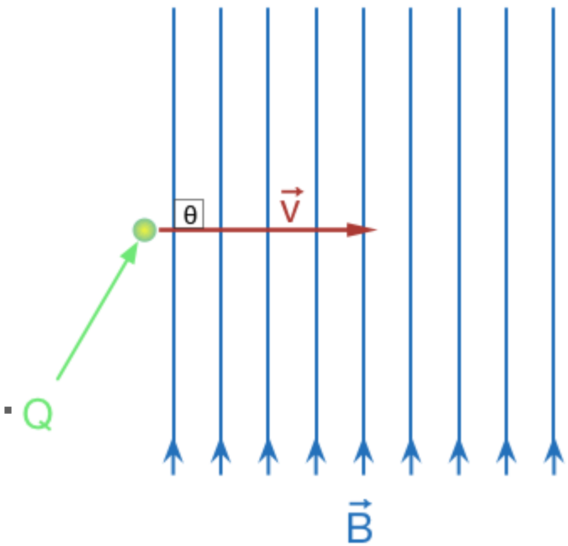
Force on a Moving Charge in a Magnetic Field

The **magnitude** of the force on a moving charge in a uniform magnetic field is given by:

$$\boxed{F = qvB\sin\theta}$$

Where:

- B is the magnetic field strength/flux density (in Tesla, T)
- q is the amount of moving charge (in Coulombs, C)
- v is the speed of the charge (in ms^{-1})
- θ is the angle between the velocity and the magnetic field lines.
  (In BP, we will focus on scenarios where $\theta = 90^0\ or\ 0^0$)

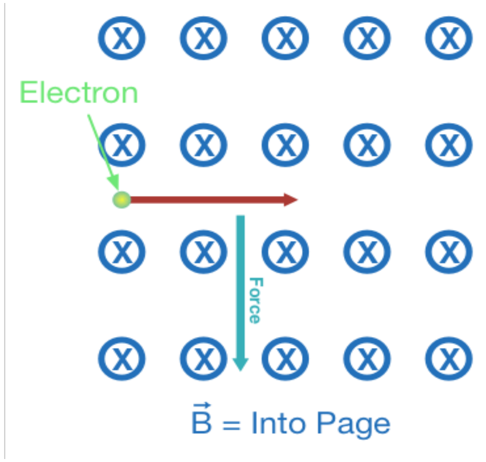


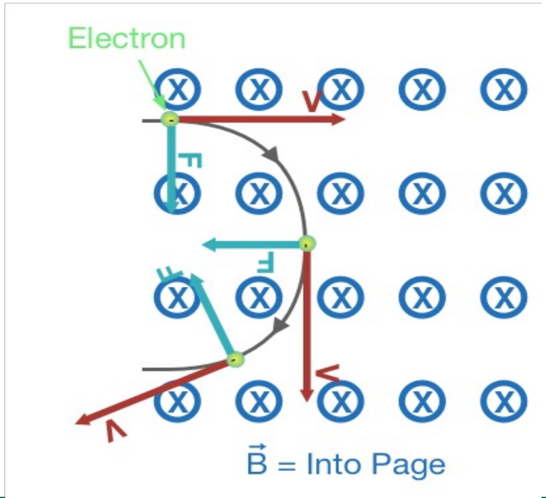
- Direction of force can be determined by **left-hand rule.**

Qn: Can you determine the direction of force acting on the positive charge above?

Circular path

- The direction of the force on a moving charge in a uniform magnetic field is found using **Left Hand Rule**.
- The direction of the force is perpendicular to the velocity. Thus the resultant motion of the moving charge is in a **circular path**.
- Qn: what is the work done on the particle by magnetic field?





Radius of Orbit, r

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- **The magnetic force acting upon a charged particle in a uniform magnetic field causes circular motion.**

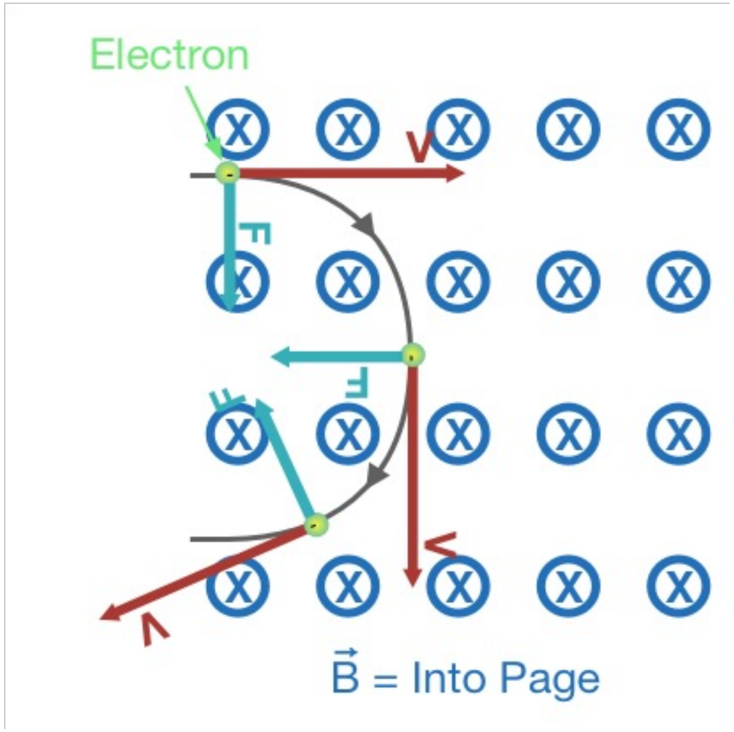
- **From the formula of centripetal force, we have**

$$qvB = m\frac{v^2}{r}$$

(v is the component of velocity perpendicular to the magnetic field)

And rearrange to find an expression for the radius of the circular path, r :

$$r = \frac{mv}{qB}$$



Use magnetic and electric together

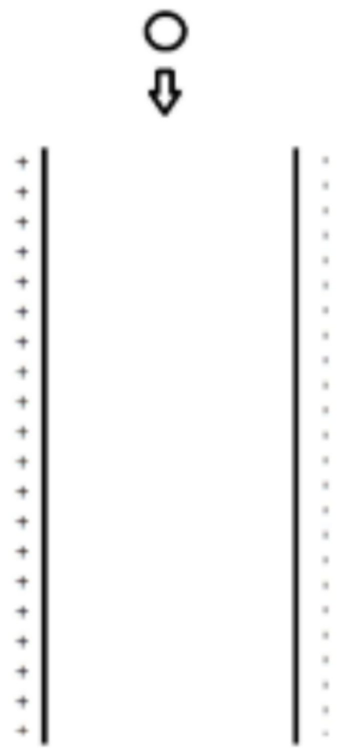
Which direction should an external magnetic field point if the proton is to move down without deflection?

1. Left	2. Right
3. In	4. Out

Use magnetic and electric together

Which direction should an external magnetic field point if the proton is to move down without deflection?

	4. Out



Use magnetic and electric together

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Magnetic fields lead to circular orbits.

Electric fields leads to parabolic paths.

When the two fields are combined (at right angles to each other), the configuration can be useful as a velocity selector of charged particles.

Qn: for the scenario at the right, if the magnitude of E field and B field are both known, how is the velocity of the particle defined?